



PROMPT ENGINEERING





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A seasoned **Global IT Trainer, Consultant, and Thought Leader** with over **25 years of experience** driving AI-powered digital transformation across industries. As the **Founder & CEO of GKTCS Innovations**, I have empowered more than **25,000 IT professionals** and collaborated with over **300 multinational corporations** to accelerate business growth through **AI-driven strategies and advanced technology adoption**.



PROMPT ENGINEERING



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Advanced Prompt Engineering & Control



Prompt strategies: zero-shot, few-shot, CoT, ReAct



Parameters: temperature, max tokens, stop sequences



Function calling & tool usage in GPT-5



Prompt templates for **project managers, developers, leaders**

Hands-On



Design structured prompts for code gen, reports, policies



Build persona-based prompts (PM vs Developer vs Leader)

Assignment

Generate
project status
updates

with structured
prompts

Agenda

Introduction to Prompt Engineering



Understanding Prompt Structure



Hands-On: Crafting Effective Prompts



Role-Based Demo – HR Team

Introduction to Prompt Engineering

What is Prompt Engineering?

Evolution with LLMs (GPT-3 → GPT-4 → GPT-5 & Multi-modal models).

Applications across HR, QA, Development, BA domains.

Prompt types: **Zero-shot, Few-shot, Chain-of-Thought.**

Understanding Prompt Structure

Key components of a prompt (Instruction, Context, Example, Constraint).

Format styles: questions, templates, structured tasks.

Prompting for **reasoning vs. factual accuracy**.

Hands-On: Crafting Effective Prompts

Activity: Basic vs. improved prompts.

Role prompting, style prompting, task chaining.

Tools: **ChatGPT, OpenAI Playground.**

Role-Based Demo – HR Team

Demo	Demo: Creating prompts for HR tasks.
Case	Case Study (HR 1): Automating job description drafting.
Case	Case Study (HR 2): Resume screening & shortlisting with ethical safeguards.

Prompt Engineering Introduction

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Prompt Engineering Introduction



Art and science of
designing



Effective
instructions
(prompts)



To get accurate,
reliable, and



useful responses
from AI systems



like ChatGPT,
Copilot, Gemini,
Claude

Prompt Engineering Introduction



Think of it as



**Asking the right
question in the right
way**

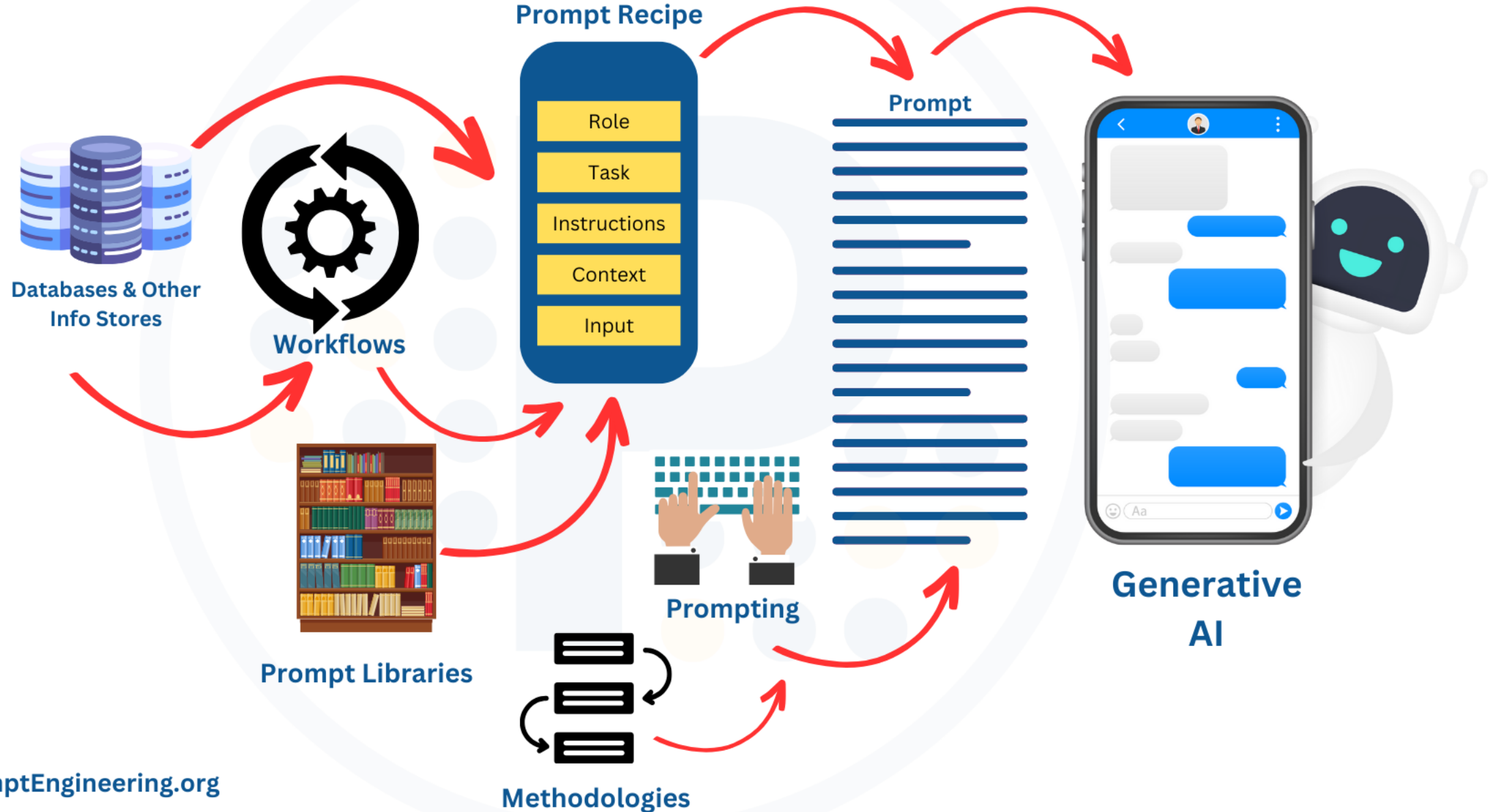


So the AI understands
your intent and



Produces relevant
outputs.

What is Prompt Engineering? Everything that goes before the prompt



Prompt Example

You are an expert in prompt engineering for large language models like ChatGPT. Create a comprehensive, step-by-step guide for beginners on how to become proficient at prompt engineering. The guide should include clear explanations, practical examples, and common mistakes to avoid. Break down the process into logical phases (e.g., understanding the model, prompt structure, refining prompts, testing, etc.). Use a friendly and instructive tone, and format the guide clearly with headings, bullet points, and examples. Conclude with actionable tips and resources for further learning.

Why this prompt works?

Role assignment:

Tells the model

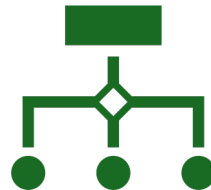
it's an expert

improves response quality.

Why this prompt works?



Clear goal:



“Create a step-by-step guide”



sets a direct objective.

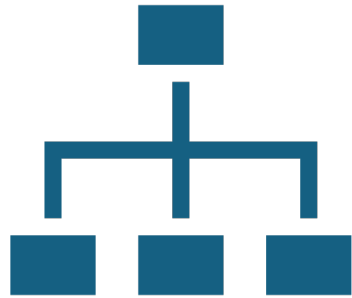
**Why this
prompt
works?**

Audience level:

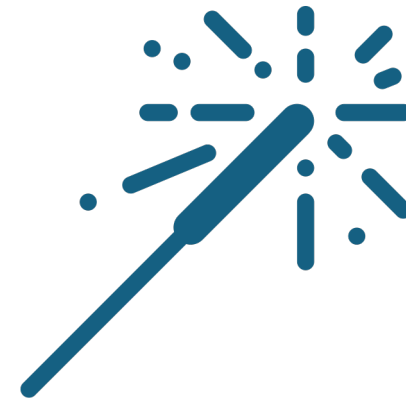
“For beginners”

**ensures explanations are
accessible.**

Why this prompt works?



Structure and formatting instructions:



Headings, bullets, examples —
this guides clarity.

Why this prompt works?



TONE GUIDANCE:



**“FRIENDLY AND
INSTRUCTIVE”**



**KEEPS IT READABLE
AND HELPFUL.**

Why this prompt works?

Extras:

Asks for mistakes to avoid +

further resources

adds practical value.

Evolution of LLMs Across Domains

Domain	GPT-3 (2020)	GPT-4 (2023)	GPT-5 (2025)	Multi-Modal Models (2024–2025)
HR	Needed detailed prompts to generate job descriptions or policy drafts.	Role-based prompting produced refined job descriptions and interview questions.	Generates entire recruitment workflows, evaluates resumes, and suggests interview panels.	Analyze CVs (PDF) + Job Descriptions (text) and auto-generate candidate match reports.

Evolution of LLMs Across Domains

Domain	GPT-3 (2020)	GPT-4 (2023)	GPT-5 (2025)	Multi-Modal Models (2024–2025)
QA	Could create test cases with very explicit instructions but often incomplete.	Generates scenario-based test cases with better coverage.	Creates full regression test suites, predicts high-risk areas, and integrates defect analysis.	Reads logs, screenshots, or video test runs and generates detailed bug reports.

Evolution of LLMs Across Domains

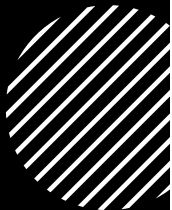
Domain	GPT-3 (2020)	GPT-4 (2023)	GPT-5 (2025)	Multi-Modal Models (2024–2025)
Development	Produced small code snippets; required significant correction.	Creates structured, context-aware code with API integration and debugging help.	Generates production-grade code, test cases, and supports repository-wide refactoring.	From diagrams (image) + requirements (text), generates backend & frontend code scaffolding.

Evolution of LLMs Across Domains

Domain	GPT-3 (2020)	GPT-4 (2023)	GPT-5 (2025)	Multi-Modal Models (2024–2025)
Business Analysis (BA)	Summarized requirements but lacked structure.	Generates structured user stories, acceptance criteria, and process flows.	Handles end-to-end BA cycle: requirements → gap analysis → recommendations.	Converts meeting transcripts (audio) + workflow diagrams (images) into formal BRDs.



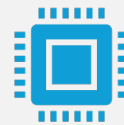
Applications Across Domains



1. Human Resources (HR)



2. Quality Assurance (QA)



3. Software Development (Dev)



4. Business Analysis (BA)

1. Human Resources (HR)



Prompt Engineering enables HR professionals



to automate and optimize daily tasks,



making people management more efficient.

HR Applications



Job
Descriptions



Resume
Screening



Interview
Questions



Policy
Drafting



Employee
Engagement

Job Descriptions



GENERATE

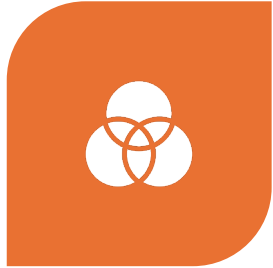


ROLE-SPECIFIC JD



IN SECONDS.

Resume Screening



COMPARE



CANDIDATE
RESUMES



WITH JD AND



HIGHLIGHT
BEST MATCHES.

Interview Questions



CREATE COMPETENCY-
BASED AND



BEHAVIORAL INTERVIEW
QUESTIONS.

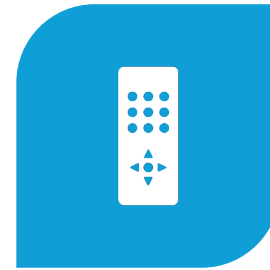
Policy Drafting



DRAFT HR
POLICIES



LEAVE POLICY,



REMOTE WORK
GUIDELINES,



PERFORMANCE
EVALUATION

Employee Engagement



GENERATE
NEWSLETTERS,



EVENT
COMMUNICATIONS, OR



FEEDBACK FORMS.

Example Prompt:

“Act as an HR Manager.

*Create an interview question set for a Data Analyst role
focusing on problem-solving and communication skills.”*

2. Quality Assurance (QA)



QA TEAMS BENEFIT
FROM



AI-DRIVEN TEST
GENERATION,



BUG DETECTION,
AND



FASTER VALIDATION
CYCLES.

2. QA Applications

**Test Case
Generation:**

Create
functional,

boundary,

negative test
cases

automatically.

2. QA Applications

**Bug
Reporting:**

Convert logs
or error
messages

into
structured
bug reports.

2. QA Applications



**TEST AUTOMATION
SUPPORT:**



GENERATE
SELENIUM,



PLAYWRIGHT,



PYTEST SCRIPTS.

2. QA Applications



Risk Coverage



Suggest



high-risk test
areas



based on
requirements.

2. QA Applications



**REGRESSION
TESTING:**



CREATE



**COMPREHENSIVE
TEST SUITES**



**FROM USER
STORIES.**

Example Prompt:

“You are a QA engineer.

*Generate 8 positive and 5 negative test cases
for an online shopping cart checkout process.”*

3. Software Development (Dev)



DEVELOPERS CAN USE
PROMPT ENGINEERING



TO ACCELERATE
CODING, DEBUGGING,



AND DOCUMENTATION.

Software Development Applications

Code Generation:

Produce code snippets,

APIs, and full modules.

Software Development Applications

Debugging:

Explain

Error messages

Suggest fixes

Software Development Applications



Documentation:



Generate



API
documentation



README files



Usage examples

Software Development Applications



Code Review:



Suggest
improvements in



Performance,



Security,



Readability.

Software Development Applications



**Architecture
Guidance:**



Draft



System Design
Patterns



Refactoring
strategies.

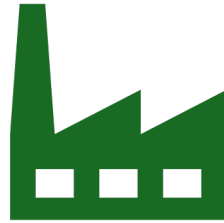
Example Prompt

“Write a FastAPI endpoint to update customer data in a PostgreSQL database, including input validation and error handling.”

4. Business Analysis (BA)



Business Analysts can
transform



raw requirements into



structured deliverables
faster.

4. Business Analysis Applications



**User Stories &
Epics:**



Generate



JIRA-style user
stories



with acceptance
criteria.

4. Business Analysis Applications

Requirement Summaries:

Convert

Meeting notes into

Structured Requirement documents.

4. Business Analysis Applications



**Process Flow
Documentation:**



Suggest



workflow
diagrams



SOPs.

4. Business Analysis Applications



Gap Analysis:



Compare



current vs.
desired



system
capabilities.

4. Business Analysis Applications



**Business Case
Creation:**



Draft



ROI analysis



Feature prioritization
documents.

Prompt Example

“You are a Business Analyst.

Convert the following notes into three user stories with acceptance criteria:

*customers should log in with email/mobile,
view order history, and download invoices.”*

Key Takeaway



Prompt Engineering isn't limited to developers



it empowers HR to manage talent,



QA to ensure quality,



Developers to code efficiently, and



BAs to structure business requirements.

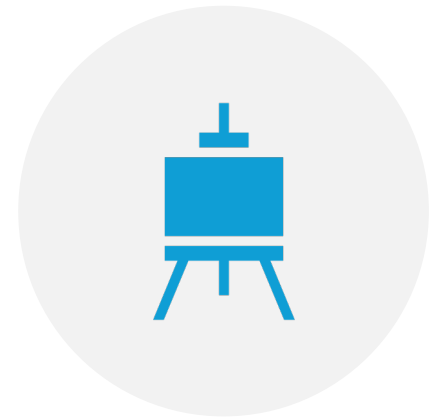
Key Takeaway



EACH DOMAIN CAN



**SAVE TIME, REDUCE ERRORS,
AND SCALE PRODUCTIVITY**



WITH EFFECTIVE PROMPT
DESIGN.

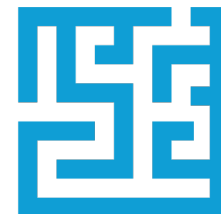
Prompt types



Zero-shot,



Few-shot,



Chain-of-Thought

Zero-Shot Prompting

Directly asking the model

to perform a task

without providing any examples.

Zero-Shot Prompting



Strength:



Fast, simple,



useful when tasks are
straightforward.

Zero-Shot Prompting



LIMITATION:



MAY LACK ACCURACY



IF THE TASK IS
COMPLEX.

Use Case (Developer – SQL query generation)

Write a SQL query to fetch all employees with salary greater than 1,00,000 from the Employee table.

Examples Across Domains

- **HR:** *“Write a job description for a Machine Learning Engineer.”*
- **QA:** *“Generate 5 test cases for login functionality.”*
- **Development:** *“Write a Python function to reverse a string.”*
- **BA:** *“Summarize requirements for an online food ordering app in bullet points.”*

2. Few-Shot Prompting

Providing

few examples in the prompt

to guide the model's response

style and format.

2. Few-Shot Prompting



Strength:



Increases accuracy and
consistency.

2. Few-Shot Prompting



Limitation:



Limited by context size;



examples must be
carefully chosen.

Use Case (Team Leader – Status updates)

Example 1:

Task: Fix login bug → Status: Completed, merged to main branch.

Example 2:

Task: Update API documentation → Status: In Progress, expected by EOD.

Now, generate status for:

Task: Implement user profile dashboard.

Examples Across Domains

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HR

*“Here are examples of interview questions:
Tell me about a time you resolved a workplace conflict.
How do you prioritize tasks under pressure?
Now generate 3 more behavioral interview questions.”*

QA

“Example test case:

Verify that valid login credentials allow access to the dashboard.

Example test case:

Verify that invalid credentials display an error message.

Generate 3 more test cases for login functionality.”

Development

*“Example code snippet:

```
# Add two numbers
```

```
def add(a, b):
```

```
    return a + b
```

Generate a function to multiply two numbers in the same style.”*

BA

Example User Story:

As a customer, I want to log in with my email so that I can access my profile.

As a customer, I want to reset my password so that I can regain access if I forget it.

Create 2 more user stories for online shopping.

Chain-of-Thought (CoT) Prompting

Asking the model to

“show its reasoning”

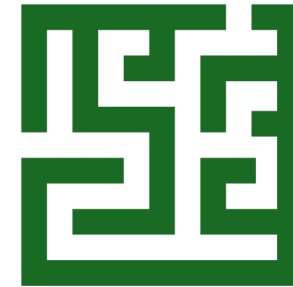
step by step

before giving the final answer.

Chain-of-Thought (CoT) Prompting



Strength:



Improves reasoning for
complex tasks.

Chain-of-Thought (CoT) Prompting



Limitation:



Longer responses,



may be slower.

Use Case (Project Manager – Effort estimation)

Estimate effort for building a feature where customers can reset their password.
Think step by step before giving the final answer.

HR

Explain step by step
how you would evaluate a resume
for a Data Scientist role,
then provide a final decision
(shortlist or reject).

QA

Think step by step.

First list possible input values for a shopping cart checkout.

Then explain expected outputs.

Finally, write test cases.

Development

Think step by step.

First explain the logic

to check if a number is prime.

Then write Python code that implements it.

BA

Step by step:

- 1) Identify stakeholders,*
- 2) Capture requirements,*
- 3) Prioritize features.*

Now convert these into 3 user stories with acceptance criteria.

ReAct (Reason + Act) Prompting



Combine **reasoning**
with **tool usage**.



Used in **Agentic AI**
systems where



GPT interacts with APIs,
databases, or tools.

Use Case (Developer – Jira + GitHub integration)

You are a project assistant.

Task: Find out if the 'Payment Gateway Timeout Bug' in Jira is resolved.

If yes, summarize the latest GitHub commit message linked to it.

Follow this format:

Thought → Action → Observation → Final Answer

GPT-5 simulates reasoning

- **Thought:** “Check Jira ticket...”
- **Action:** Call Jira API (in real agent setup).
- **Observation:** “Bug marked resolved.”
- **Final Answer:** “Commit message: Fixed payment timeout by adding retry logic.”

Key Insights

Zero-Shot → Use when task is simple
(SQL, summaries, translations).

Few-Shot → Use for **consistent formatting**
(Jira, test cases, reports).

Key Insights

CoT → Use for **complex reasoning**

(effort estimation, root cause analysis).

ReAct → Use in **AI agents** that integrate with tools

(Jira, GitHub, APIs).

Key GPT-5 Parameters

1. temperature
2. max_tokens
3. stop sequences

temperature

- Controls **creativity vs. determinism**.
- Range: **0 → 2**
- 0 = deterministic (same answer each time, safe for coding/testing).
- 1 = balanced (default, natural variety).
- >1 = very creative (good for brainstorming).

max_tokens

- Controls **how long** the response can be.
- Useful to prevent **very long answers**.
- Think of tokens $\approx$ words/parts of words (1 token $\approx$ 4 chars in English).

stop sequences

- Defines **where the model should stop generating**.
- Useful to **enforce formatting** (tables, JSON, dialogues).
- Example: stop when model outputs "\nHuman:"
- so it doesn't continue a dialogue.

Practical Takeaways

- **temperature** → Controls creativity (coding = 0, brainstorming = high).
- **max_tokens** → Controls response length (short answers vs. detailed reports).
- **stop** → Enforces clean formatting in outputs (chatbots, JSON, reports).

Function Calling & Tool Usage in GPT-5

What is Function Calling?

Function calling allows GPT-5 to **decide when and how to call external functions/tools.**

Useful for

- Fetching live data (APIs, DB queries).
- Enforcing structured outputs (JSON).
- Agentic AI workflows (Reason + Act).

Why This Matters ?

Developers

Can integrate GPT-5 with
company APIs
(Jira, GitHub, Databases).

Why This Matters ?

Team Leaders

Can build AI copilots

that **automate daily standup summaries,**

bug tracking,

testing workflows.

Why This Matters ?

Project Managers

Get **real-time insights**

(status reports, estimations, risk checks)

directly from AI assistants.

Prompt templates

Write a **clear instruction + context + constraints/output format.**

Add **variables** like

{project}, {owner}, {deadline}.

Prompt templates

Keep **role & audience** explicit

(“You are a senior PM...”).

Prefer **structured outputs**

(JSON, tables) for tools and dashboards.

Prompt templates

They use Python `str.format` placeholders like `{project}`.

Change to your templating style if needed.

Project Managers

1) Daily Status (JSON, concise)

You are a senior Agile Project Manager.

Summarize the current status for project

"{project}" (sprint: {sprint}) using ONLY valid JSON.

Project Managers

Inputs:

- Objectives: {objectives}
- Done since last update: {done}
- Risks/issues: {risks}
- Upcoming milestones (with dates): {milestones}

Project Managers

Rules:

- Be factual and brief.
- No extra prose outside JSON.

Project Managers

Output JSON schema:

```
{  
  "project": string,  
  "sprint": string,  
  "overall_status": "On Track" | "At Risk" | "Off Track",  
  "highlights": [string],  
  "risks": [{"id": string, "desc": string, "impact": "low|med|high", "owner":  
string, "mitigation": string}],  
  "blockers": [string],  
  "next_7_days": [string]  
}
```

Developers

Bug → Minimal Repro + Fix Plan

You are a senior Python developer.

Bug report:

{bug_report}

Developers

Tasks:

- 1) Produce a minimal reproducible example (MRE).
- 2) Hypothesize root cause(s) with confidence levels.
- 3) Propose a fix plan with steps, unit tests, and rollback plan.
- 4) Output sections titled: MRE, RootCause, FixPlan, Tests, Rollback.

Pro tips

Add constraints:

“only valid JSON”, “no prose”, “max 5 bullets”.

Set temperature:

0–0.3 for precision (reports/code), 0.7+ for ideation.

Use stop sequences for dialogue formats

(e.g., stop=["Human:"]).

Pro tips

Few-shot for style consistency

(paste 1–2 short examples above your task).

ReAct when you'll wire tools

(Jira, GitHub, DB)—it scales to agents.

Understanding Prompt Structure

- Key components of a prompt
- (Instruction, Context, Example, Constraint).
- Format styles: questions, templates, structured tasks.
- Prompting for **reasoning vs. factual accuracy**.

Key Components of a Prompt

- Instruction
- Context
- Example(s)
- Constraint(s)

Key Components of a Prompt

- **Instruction**
- The direct task you want the AI to perform.
- Clear, action-oriented wording.

Key Components of a Prompt

Context

Background information

to help the AI understand the situation.

Provides relevance and domain alignment.

Key Components of a Prompt

Example(s)

Demonstrations of the desired style,
format, or structure.

Helps guide the AI's output.

Key Components of a Prompt

Constraint(s)

Rules or limitations for the output.

Examples: word limits, tone, format, or specific inclusions/exclusions.

Human Resources (HR)

- **Instruction:** *“Write an interview question set.”*
- **Context:** *“The role is for a Data Analyst in the BFSI domain.”*
- **Example:** *“Example question: How would you validate financial datasets for accuracy?”*
- **Constraint:** *“Provide 5 questions, each focusing on problem-solving skills.”*

HR Prompt Example

Write 5 interview questions for a
Data Analyst in the BFSI domain.

Example:

How would you validate financial datasets for accuracy?

Keep the questions problem-solving oriented.

Quality Assurance (QA)

- **Instruction:** *“Generate test cases.”*
- **Context:** *“The application is an e-commerce checkout system.”*
- **Example:** *“Example test case: Verify that valid credit card details allow successful payment.”*
- **Constraint:** *“Provide at least 3 positive and 3 negative test cases.”*

Full Prompt

Generate test cases for
an e-commerce checkout system.

Example:

Verify that valid credit card details allow successful payment.
Provide at least 3 positive and 3 negative cases.

Development (Dev)

- **Instruction:** *“Write a Python function.”*
- **Context:** *“The function should interact with a PostgreSQL database for customer data.”*
- **Example:** *“Example: A function to fetch customer by ID with error handling.”*
- **Constraint:** *“Use FastAPI framework and include docstrings.”*

Full Prompt

Write a FastAPI endpoint in Python
that updates customer data in a PostgreSQL database.

Example:

fetching customer by ID with error handling.

Use FastAPI and include docstrings.

Full Prompt

Create 3 user stories for an online food delivery app.

Example:

As a customer, I want to log in with my email so that I can view my past orders.

Add acceptance criteria in Gherkin format.

Key Takeaway

- Every effective prompt is made of:
- **Instruction → What to do**
- **Context → Where/why it applies**
- **Example → How the output should look**
- **Constraint → Rules to follow**

Hands-On Step by Step: Crafting Effective Prompts

- Activity: Basic vs. improved prompts.
- Role prompting, style prompting, task chaining.
- Tools: **ChatGPT, OpenAI Playground.**
- https://platform.openai.com/chat/edit?prompt=prompt_68af3c8d896481908f1db839ddc561c90db33e8a1d08c679&version=1

ChatGPT

Use a fresh chat for each activity.

Start with a short “system instruction” like:

“You are a precise, enterprise-grade assistant.

Prefer structured, verifiable outputs.”

OpenAI Playground

Pick a chat-capable model.

Set a **System** message for role,
adjust **Temperature**

(0.2–0.5 for accuracy; 0.7–1.0 for creativity), and
keep Max Output high enough for your task.

The Playground lets you experiment with roles
and parameters interactively.

Further Questions



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Happy Learning@!!
Thanks for Your
Patience 😊

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